THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** the mass of the ball and its speed and direction	these decide whether the ball crosses the boundary
2	**(d)** it comes from the Latin word celeritas meaning speed	symbols come from history and agreement
3	**(c)** describes a regular pattern observed in nature	laws describe what happens
4	**(a)** a broad idea that helps make sense of many situations	principles are broad ideas
5	**(a)** tiny differences in conditions grow over time	sensitivity to initial conditions
6	**(b)** 15 %	$\frac{6}{40} \times 100 = 15$
7	**(b)** 4 %	$\frac{2}{50} \times 100 = 4$
8	**(c)** increases with temperature	as per solubility curves
9	**(a)** larger, well-formed crystals	particles get time to arrange
10	**(b)** a separating funnel	immiscible liquids form two layers

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Re-examine the assumptions, the model or the measurements instead of rejecting evidence (2).	2
12	e.g. What was the humidity the last time it rained? Is the temperature dropping now? What are today's wind speed and direction? (1+1).	1+1
13	Fractional distillation (1); the components have small differences (< 25 °C) in boiling points (1).	1+1
14	Pencil graphite does not dissolve in the solvent and does not move with it, so it will not interfere with the separation (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Balloons per minute = 3, litres per minute = 6; per day $6 \times 1440 = 8640$ litres (2), close to 10 000 L so the estimate is reasonable (1).	2+1
16	The mixture stands to form two layers (1); the lower layer is run out through the stopcock, the interface portion is collected separately, then the upper layer (1); based on immiscibility and difference in density (1).	1+1+1
17	Daily need per adult ≈ 2000–2500 kcal (1); find kcal in 100 g of cooked rice and compute the daily and monthly requirement (1); check that the answer is sensible: 100 g is too little, a few tonnes too much (1).	1+1+1
	OR: Divisions are made by us to organise knowledge (1); real problems such as climate change and medicine need ideas from several branches (2).	1+2
18	Sublimation: solid → vapour directly on gentle heating (1); camphor deposits as solid on the cool funnel wall (deposition) (1); sand remains (1).	1+1+1
	OR: Solution: smallest particles, no settling, no scattering; colloid: intermediate, no settling, scatters light; suspension: large, settles, scatters/opaque (1 each).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) To anticipate outcomes before or without experiments (1). (ii) Correct: confidence in the theory grows (1); failed: assumptions, models or measurements are re-examined (1). (iii) Distance of a kicked football; CO_2 produced in baking; breathing rate while running (2).	1+2+2
	OR: (i) A shared precise language allows results to be compared; maths expresses relationships (2). (ii) m — mass, v — velocity, F — force, I — current (any three) (2). (iii) From Latin celeritas meaning speed (1).	2+2+1

20	(i)–(ii) Magnet removes iron (magnetic property); heat gently with an inverted funnel to sublime camphor; add water, filter to remove sand (insoluble); evaporate to get salt (3). (iii) Evaporation/crystallization (1). (iv) Sublimation (1).	3+1+1
	OR: (i) Direct change from solid to vapour on heating; camphor, naphthalene, dry ice, ammonium chloride (2). (ii) Heat the mixture in a china dish covered with an inverted funnel with cotton plug; camphor deposits on the funnel's inner wall, sand remains (2). (iii) Deposition (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10 000 litres	0.5 × 20 000
2	**(b)** Using the speed and angle of a kick to find how far a football will travel	reasoned expectation from evidence
3	**(b)** direction — choosing models and the right questions	exploration needs direction
4	**(b)** distance and average speed	the details that decide the time
5	**(a)** use relationships and check with an estimate	reasoning first, calculation second
6	**(b)** solid particles in a gas	solid dispersed in air
7	**(d)** they are too small	size less than about 1 nm
8	**(b)** oil and water	form separate layers
9	**(d)** distillate	collected in a separate vessel
10	**(b)** soda water (CO_2 in water)	uniform composition throughout

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Weather depends on many changing factors and tiny differences grow over time, giving very different outcomes (2).	2
12	It helps check whether a calculated answer is reasonable, build intuition and detect errors (2).	2
13	Use a separating funnel: the denser water layer is run out through the stopcock first and then the oil (2).	2
14	Camphor changes directly from solid to vapour on heating and condenses back as a solid, while sand does not sublime (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Physics: how particles move and are filtered; chemistry: fibres and materials; biology: size and behaviour of germs (3).	1+1+1
16	Heterogeneous but appear homogeneous; particles do not settle; scatter light (Tyndall effect); pass through filter paper (any three) (2); e.g. milk, fog, smoke (1).	2+1
17	An equation links quantities, e.g. speed = distance/time relates three quantities (2); it lets us predict where an object will be later (1).	2+1
	OR: Understanding the situation and the relevant quantities comes first (1); a rough estimate can reveal whether a result is possible or absurd (1); exact numbers without reasoning can still be wrong (1).	1+1+1
18	Milk particles are colloidal and scatter light (2); salt solution particles are too small to scatter (1).	2+1
	OR: Evaporation removes the solvent completely and may leave impurities (1.5); crystallization uses cooling of a hot saturated solution and gives pure crystals (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Many complex processes inside the star (1). (ii) Temperature, pressure and how atoms formed ions (1). (iii) A simplified representation retaining the key factors (2). (iv) Science often begins by ignoring details (1).	1+1+2+1
	OR: (i) They describe patterns and explain causes so that new situations can be anticipated (2). (ii) Measurements of temperature, pressure, humidity and wind in models give forecasts (1). (iii) Tiny changes grow with time; models ignore some details so long-range forecasts are uncertain (2).	2+1+2
20	(i) Oil–water: heterogeneous; alcohol–water: homogeneous (1). (ii) Separating funnel (1). (iii) Distillation since boiling points differ (2). (iv) Alcohol and water do not form layers (1).	1+1+2+1

	OR: (i) A homogeneous mixture of metals or a metal and non-metal (1). (ii) Melt the components together and cool the mixture (1). (iii) Alloys are stronger, harder or more resistant to corrosion (2). (iv) Copper and zinc (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a single point	size and shape are ignored
2	**(b)** 299792458 m/s	a defined physical constant
3	**(c)** provides a tested explanation of why patterns occur	theory is evidence-based, not a hunch
4	**(c)** may be improved or revised when new evidence appears	science is open to correction
5	**(a)** ask what mechanism and measurable change could make it harmful	test claims with questions and evidence
6	**(a)** 12 %	$\frac{12}{100} \times 100 = 12$
7	**(b)** 9 %	$\frac{18}{200} \times 100 = 9$
8	**(b)** decreases with an increase in temperature	why a warm soda goes flat
9	**(b)** simple distillation	e.g. acetone and water
10	**(c)** sublimation	camphor sublimes, sand does not

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$14 \times 60 \times 24 = 20160$ — about 20 thousand breaths (2).	1+1
12	Physics: pressure–temperature relation; chemistry: cooking changes food chemically (or biology: nutrients) (1+1).	1+1
13	Solution: homogeneous, particles do not settle, pass through filter paper; suspension: heterogeneous, particles settle on standing, retained by filter paper (1+1).	1+1
14	Scattering of a light beam by particles in a colloid, making its path visible (1); e.g. a torch beam in fog or in milk and water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repeated observations reveal patterns (1); refined measurements improve accuracy (1); together they allow laws and theories to be tested (1).	1+1+1
16	ORS needs specified amounts of salt and sugar in a fixed amount of water (1.5); pesticide: too little will not protect crops, too much damages them (1.5).	1½+1½
17	Using motion to predict how far a football travels (1); using chemistry to estimate CO_2 produced when baking (1); using biology to predict breathing during exercise (1) (any two with reasoning).	1+1+1
	OR: It is based on belief, not on measurable evidence or past patterns (1); a scientific prediction uses reasoned expectations from data (1); it can be tested and refined (1).	1+1+1
18	The solubility of gas (CO_2) in liquids decreases with temperature (2), so the gas escapes (1).	2+1
	OR: Homogeneous mixtures of metals/metal and non-metal (1); they have properties better than the pure metal (hardness, resistance) (1); brass, bronze (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Ask for evidence, measurements and mechanisms (1). (ii) ‘Food becomes harmful in an eclipse’ — what measurable change occurs? (2). (iii) Understand the situation, identify quantities, estimate (1). (iv) Testing improves ideas; no theory is final (1).	1+2+1+1
	OR: (i) Climate change (physics, chemistry, biology, earth science) and mask design during the pandemic (2). (ii) Physics: circuits and waves; chemistry: battery materials; biology: effects on health (2). (iii) Helps understand the world fully and solve problems creatively (1).	2+2+1

20	(i) Fish and other aquatic animals breathe oxygen dissolved in water (1). (ii) It decreases (1). (iii) Solubility of gases falls as temperature rises; so warm water holds less oxygen and life is affected (3).	1+1+3
	OR: (i) Wrong proportions may not rehydrate properly or may harm the patient (1). (ii) Sugar $\frac{20}{1000} \times 100 = 2\%$; salt 0.35% (2). (iii) Volume of liquids is easier to measure than mass (2).	1+2+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** mass and velocity	symbols stand for quantities
2	**(a)** several branches of science together	real-world problems cross disciplinary boundaries
3	**(b)** it is far too little for a whole month	sense check
4	**(b)** a smooth sphere layered into distinct regions	a model of layers
5	**(c)** local objects or opinions	fairness in trade and research
6	**(b)** tiny liquid droplets in air	liquid dispersed in gas
7	**(b)** suspension	particles settle on standing
8	**(a)** homogeneous	uniform composition
9	**(b)** a beam of light passing through fog	colloidal particles scatter light
10	**(d)** large enough to be seen and to settle	particles settle on standing

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It expresses relationships between quantities compactly and allows them to be tested carefully (2).	2
12	Approximate: estimating the rice needed for a month. Exact: the dose of a medicine (1+1).	1+1
13	A homogeneous mixture of two or more metals, or a metal and a non-metal (1); e.g. brass (copper + zinc) (1).	1+1
14	Smoke (solid in gas), fog (liquid droplets in gas), dust in air (any two) (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Understand the situation first (1); identify the quantities that matter (1); make a rough estimate to check the answer (1).	1+1+1
16	Magnet removes iron filings (1); add water, filter to remove sand (1); evaporate the filtrate to obtain salt (1).	1+1+1
17	So that observations and results are communicated clearly across the world (1); e.g. ‘cell’ in biology means a unit of life, not a prison room or a battery (2).	1+2
	OR: Local objects differ from place to place and opinions vary (1); standards make results comparable and trade fair (1); e.g. a kilogram of rice is the same everywhere (1).	1+1+1
18	Layers form because of a density difference; with equal densities there is no force separating them, so they stay mixed as droplets (3).	1+1+1
	OR: Gas particles move freely in all directions and mix uniformly (1.5); example: air or H_2 + O_2; heterogeneous: smoke/fog (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Understand the situation; identify quantities; use relationships; estimate to check (2). (ii) ≈ 2 min/page ⇒ 400 min ≈ 6–7 hours (2). (iii) Compare with known cases: a novel takes a few hours, so the estimate is reasonable (1).	2+2+1
	OR: (i) Physics: car as a point; chemistry: atoms as spheres; biology: labelled cell diagram; earth science: Earth as a layered sphere (2). (ii) Size and shape; electron cloud details; unimportant organelles; mountains and valleys (2). (iii) The model chosen depends on the question being asked (1).	2+2+1
20	(i) Evaporation removes the solvent; crystallization gives pure solid on cooling a saturated solution; distillation separates liquids by boiling points and recovers the solvent (3). (ii) Salt from salt water; pure copper sulfate; acetone from water (2).	3+2
	OR: (i) Suspension and colloids (1). (ii) Gas dispersed: foam; liquid dispersed: fog, milk; solid dispersed: smoke, muddy water (2). (iii) Colloids (fog, milk, smoke) show the effect; true solutions and settling suspensions do not clearly (2).	1+2+2

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